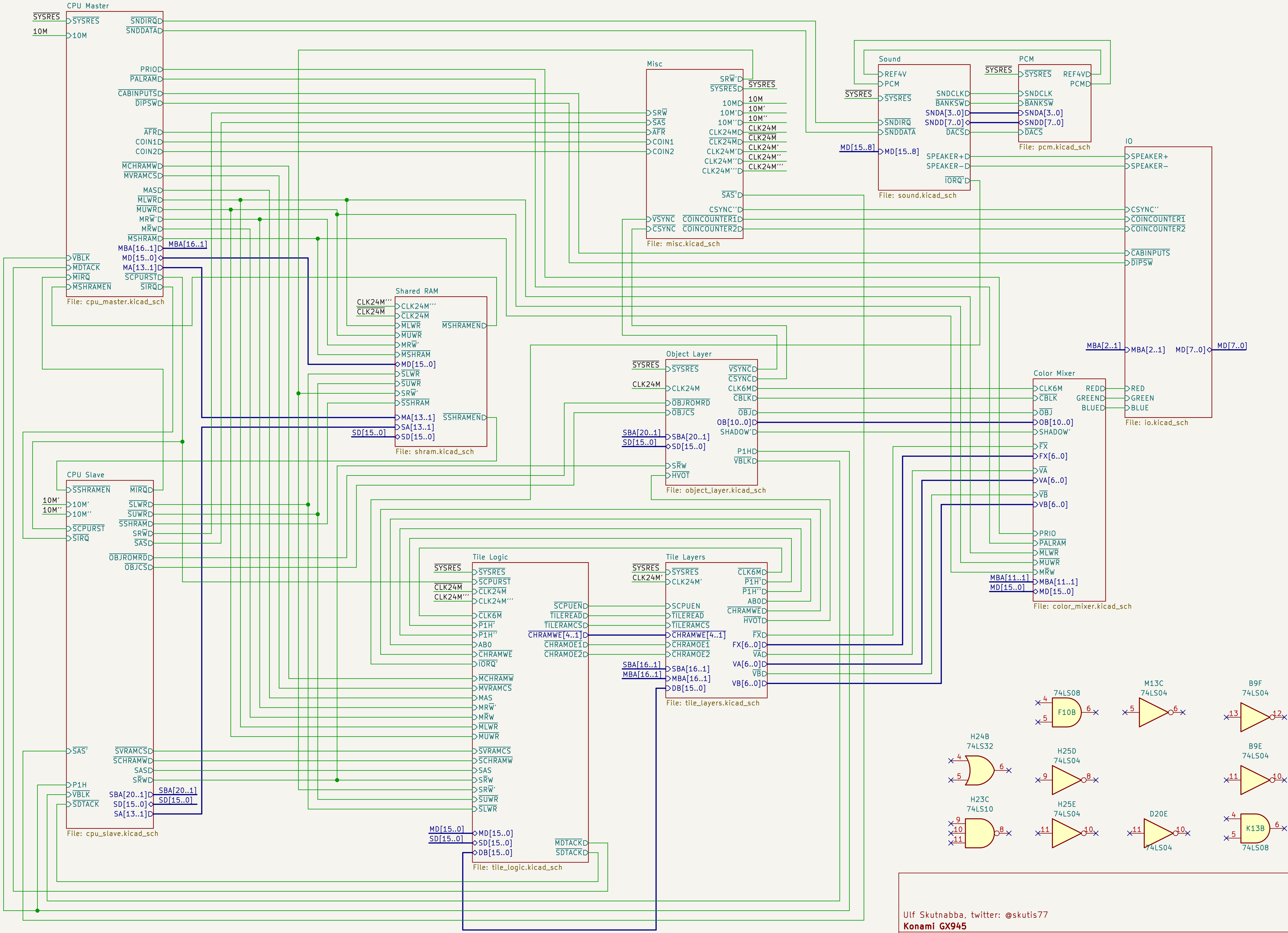

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Konami GX945

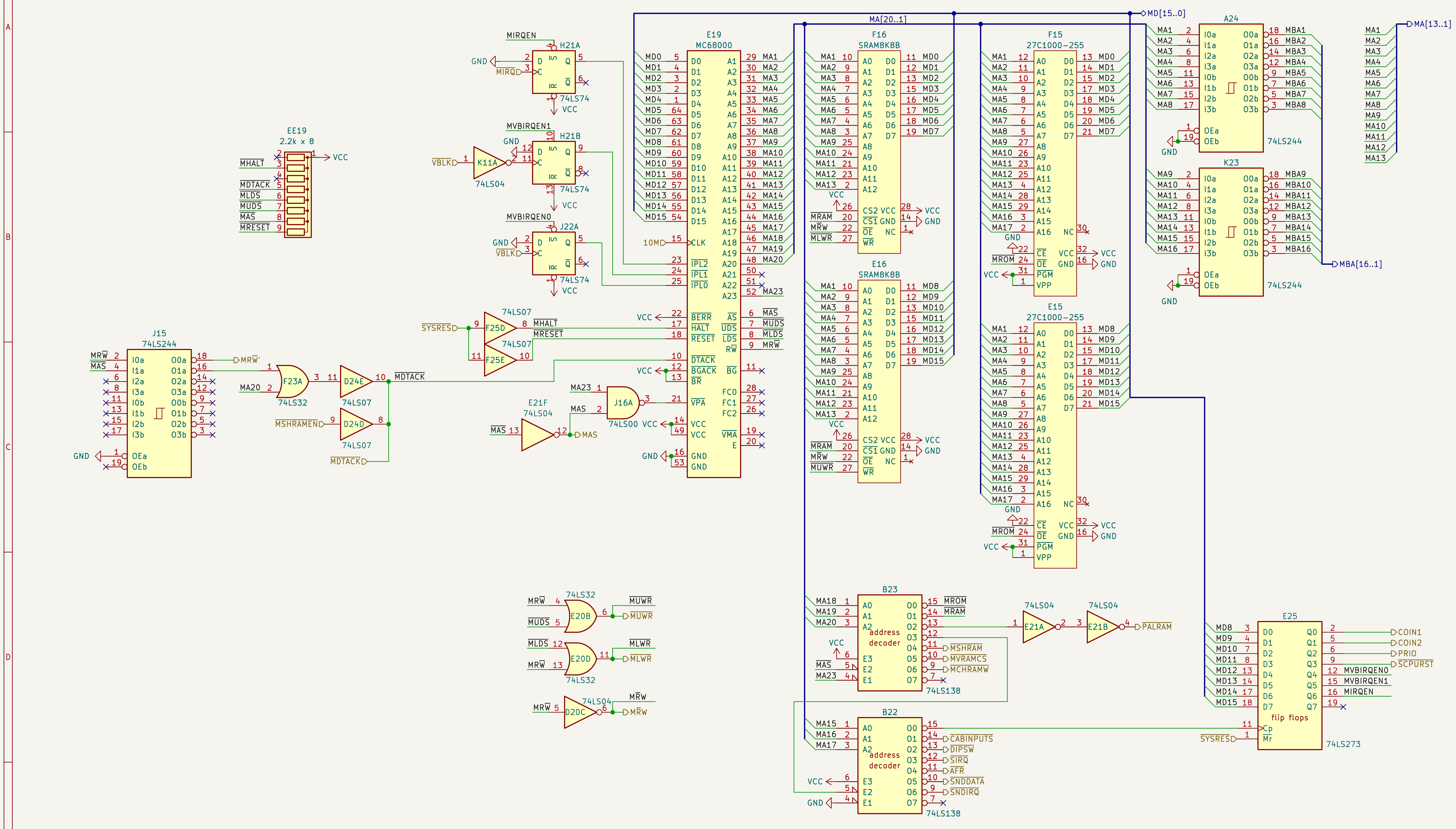
Sheet: /

File: grad3.kicad_sch

Title: Gradius 3

Size: A3 Date: 2026-05-10 Rev:

KiCad E.D.A. 8.0.9 Id: 1/12



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Konami GX945

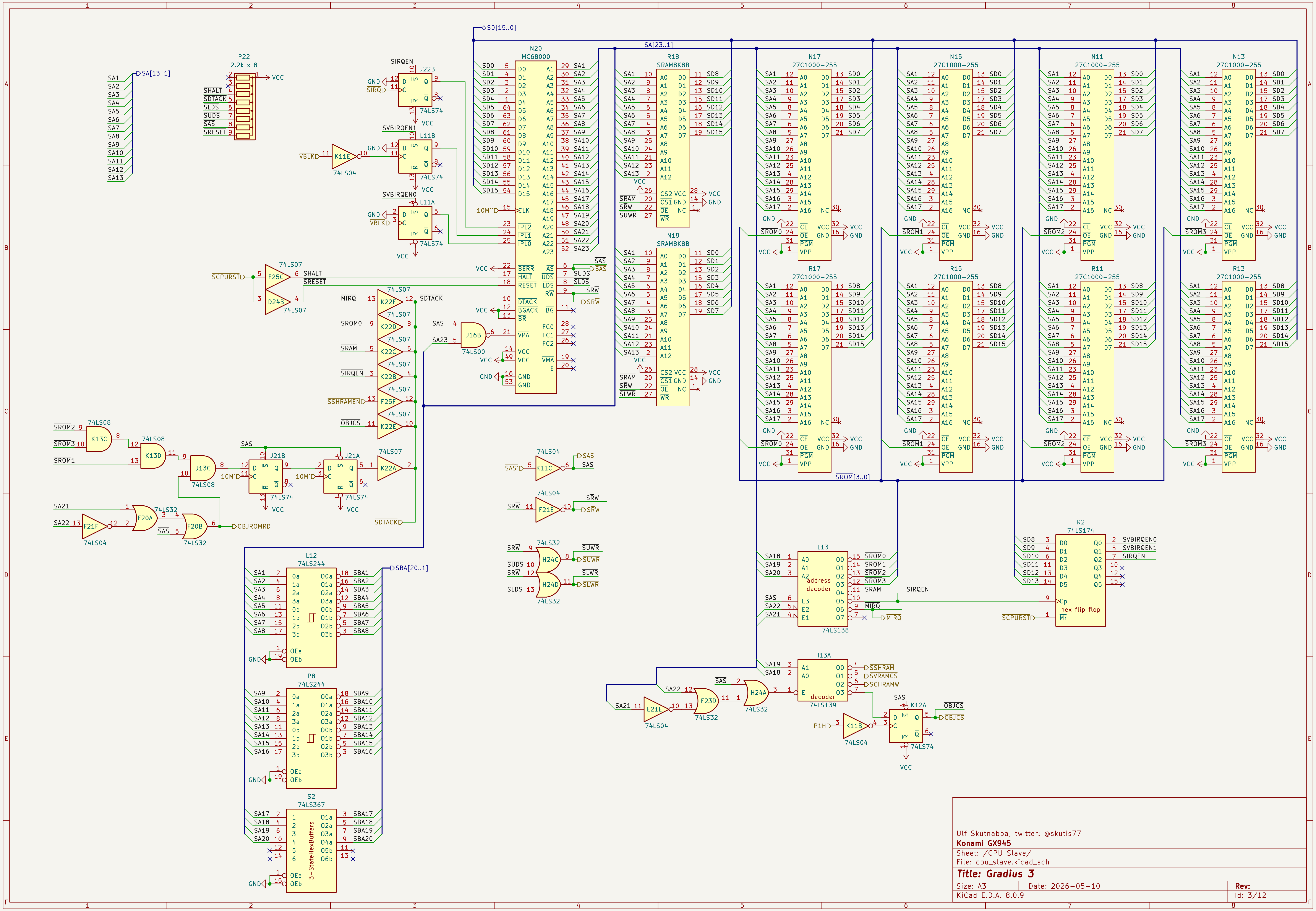
Sheet: /CPU Master/

File: cpu_master.kicad_sch

Title: Gradius 3

Size: A3 | Date: 2026-05-10 | **Rev:**

KiCad E.D.A. 8.0.9 | Id: 2/12



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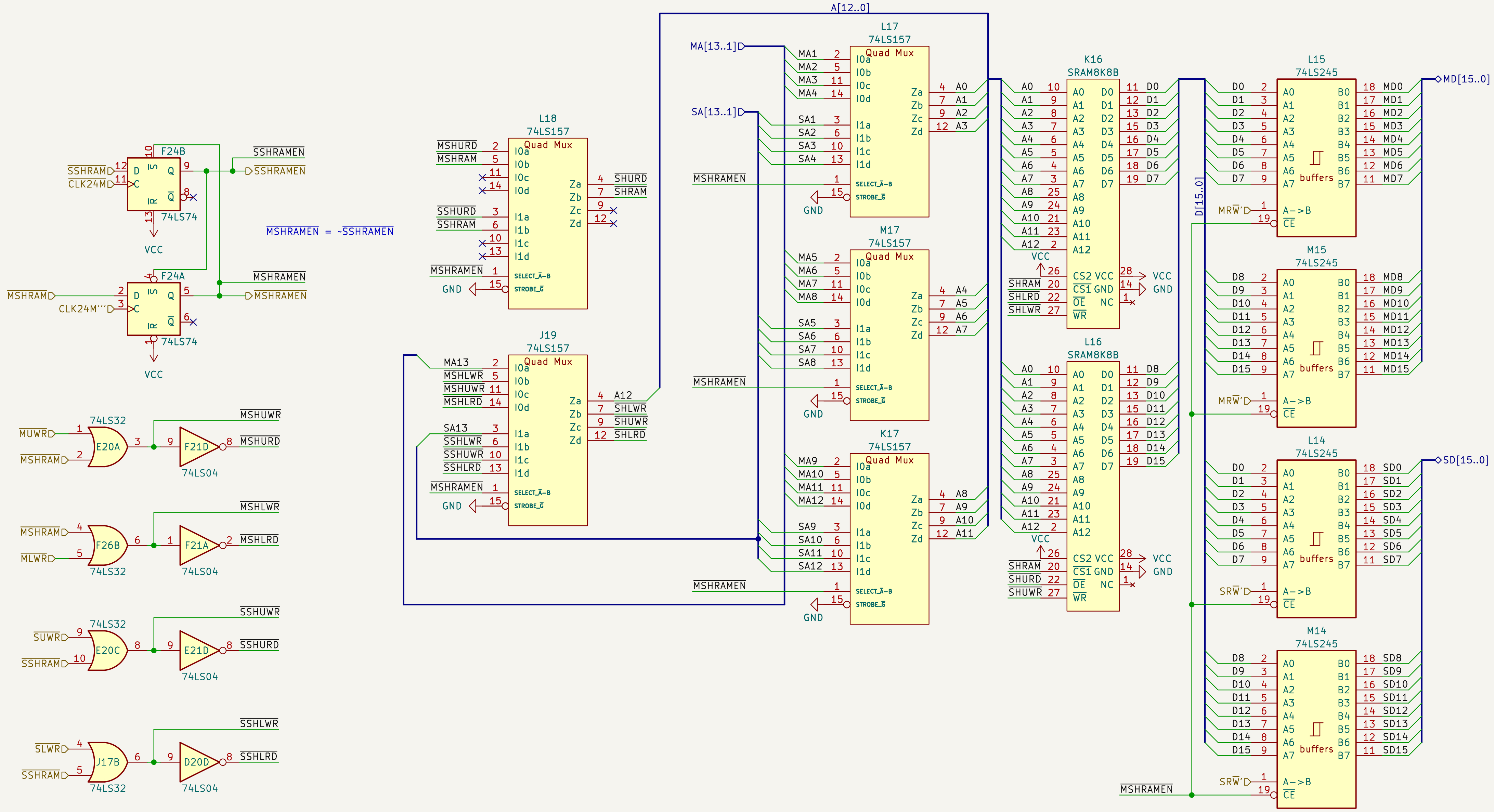
Sheet: /CPU Slave/

File: cpu_slave.kicad_sch

Title: Gradius 3

Size: A3 | Date: 2026-05-10 | Rev:

KiCad E.D.A. 8.0.9 | Id: 3/12



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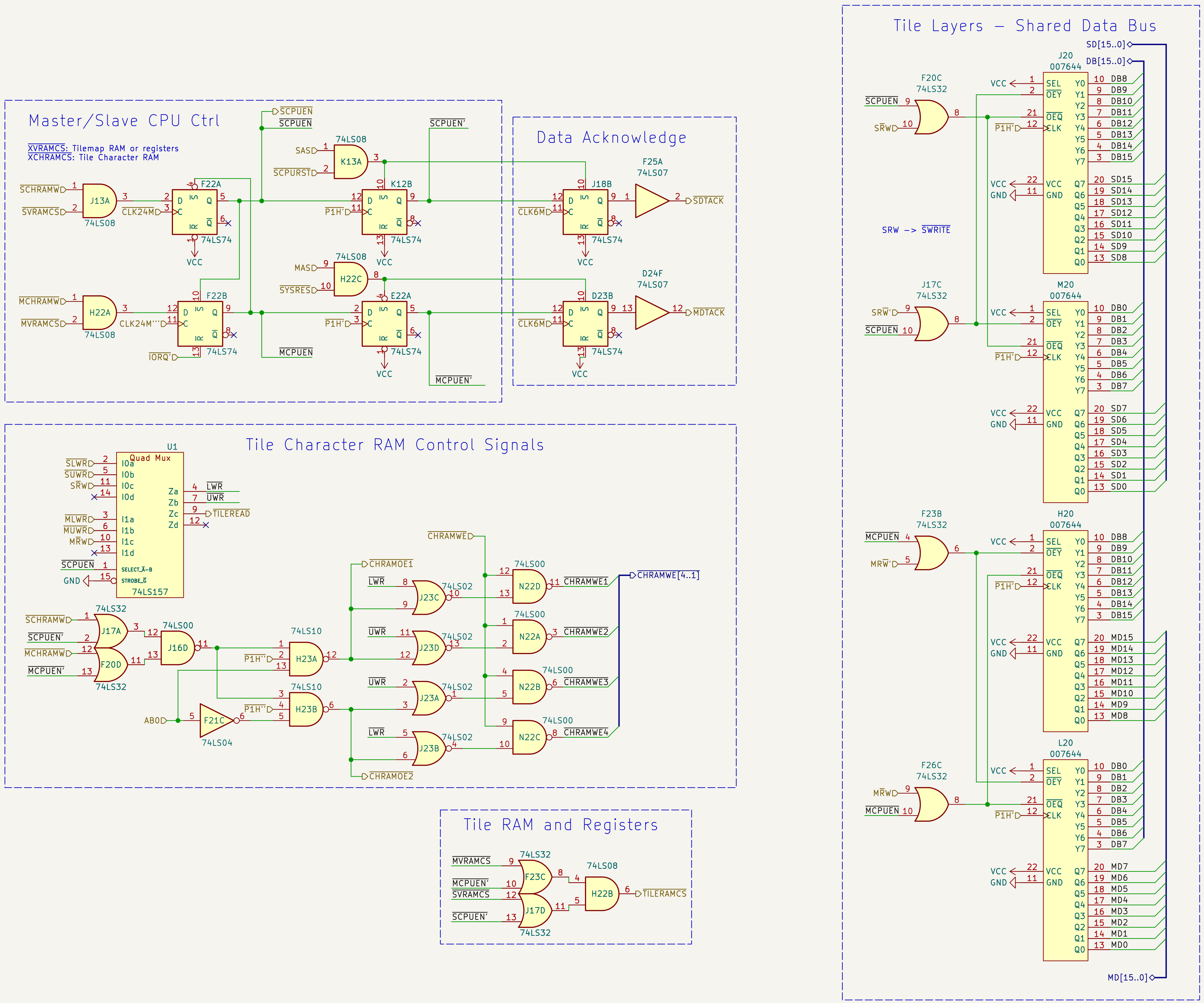
Sheet: /Shared RAM/
File: shram.kicad_sch
Title: Gradius 3
Size: A3 Date: 2026-05-10 Rev:
KiCad E.D.A. 8.0.9 Id: 4/12



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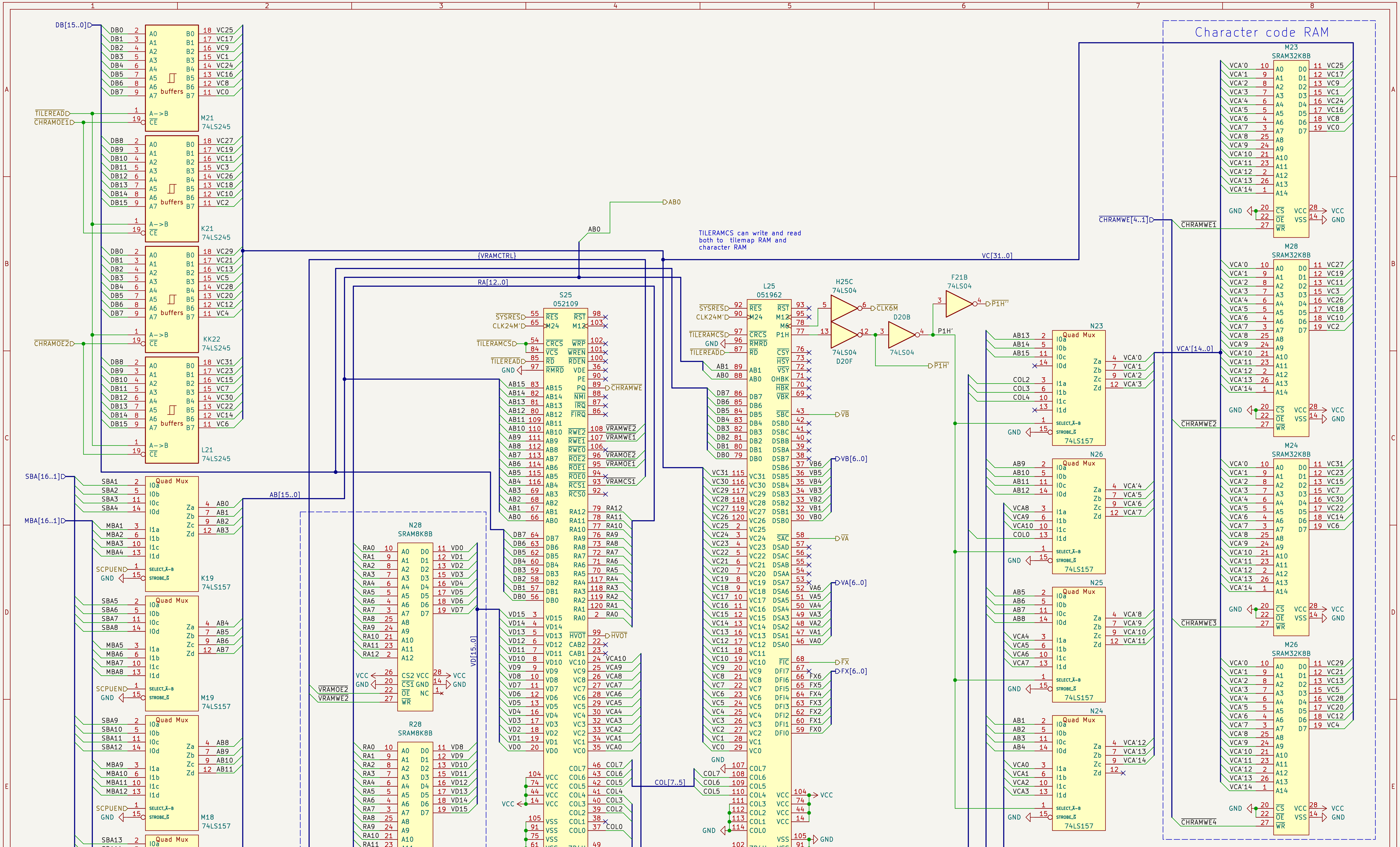
Konami GX945

Sheet: /Tile Logic/

File: tile_logic.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	Rev:
KiCad E.D.A. 8.0.9		Id: 5/12



Ulf Skutnabba, twitter: @skutis77

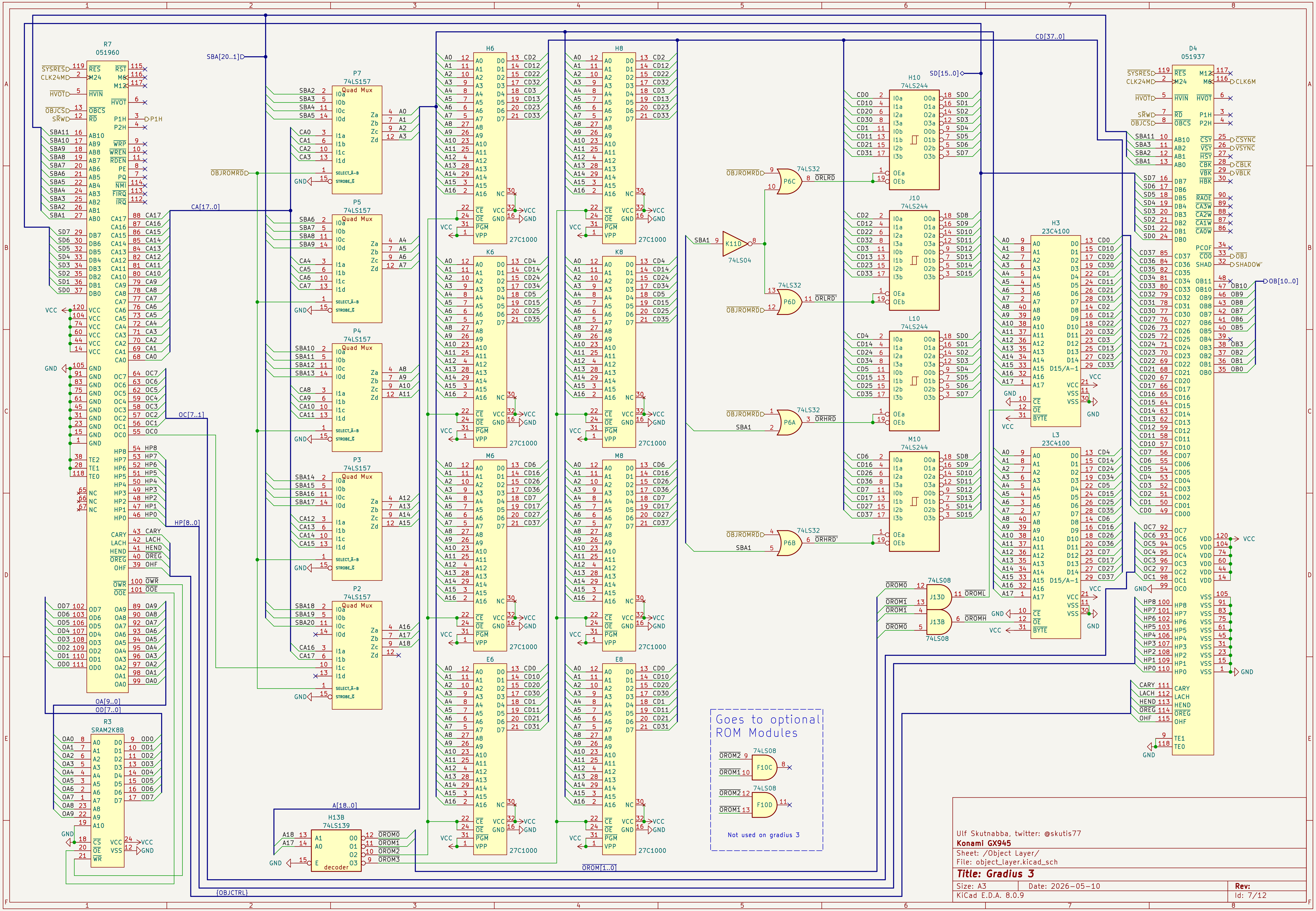
Konami GX945

Sheet: /Tile Layers/

File: tile_layers.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	Rev:
KiCad E.D.A. 8.0.9		Id: 6/12



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Konami GX945

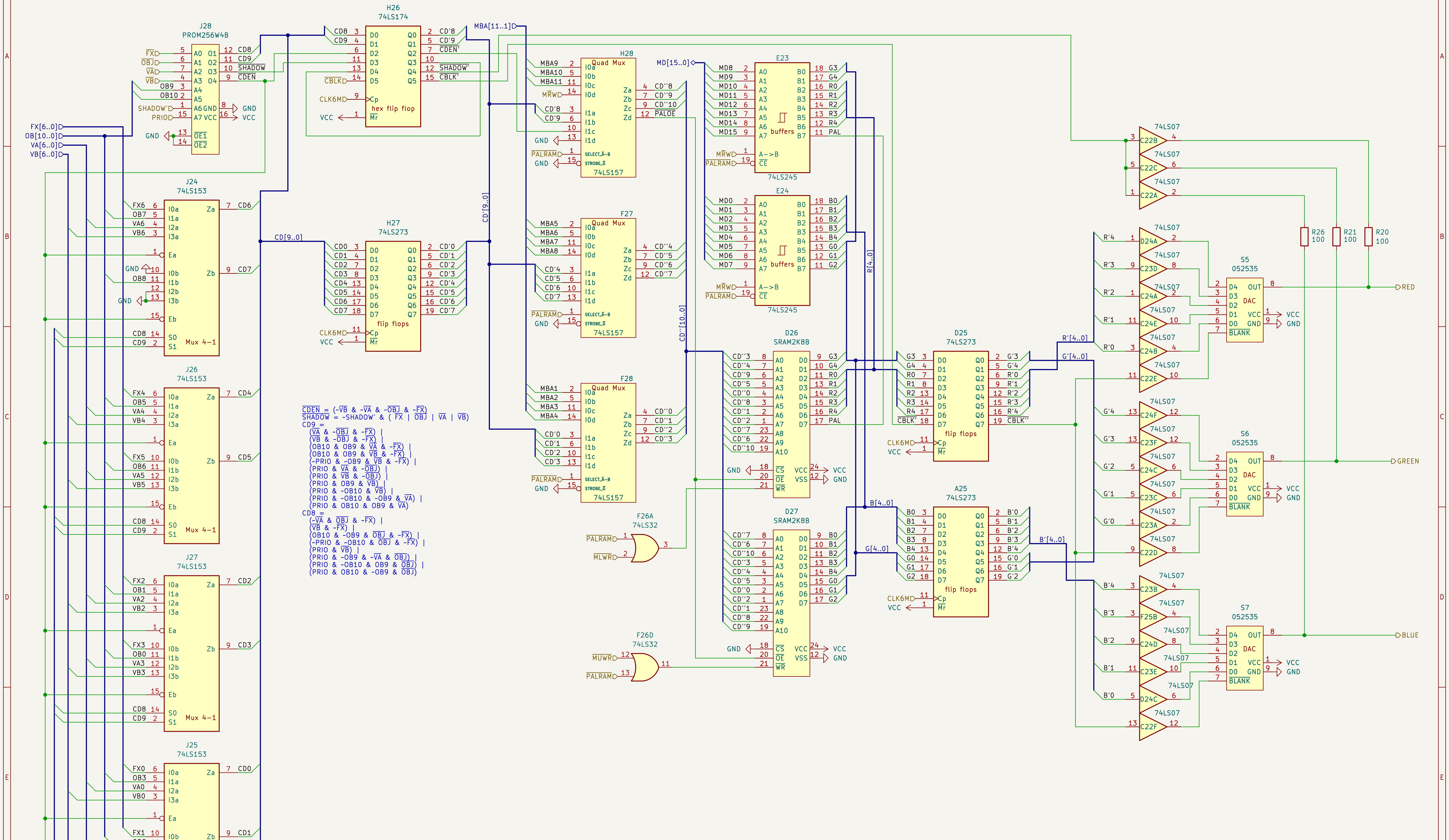
Sheet: /Object Layer/

File: object_layer.kicad_sch

Title: Gradius 3

Size: A3 | Date: 2026-05-10 | Rev:

KiCad E.D.A. 8.0.9 | Id: 7/12



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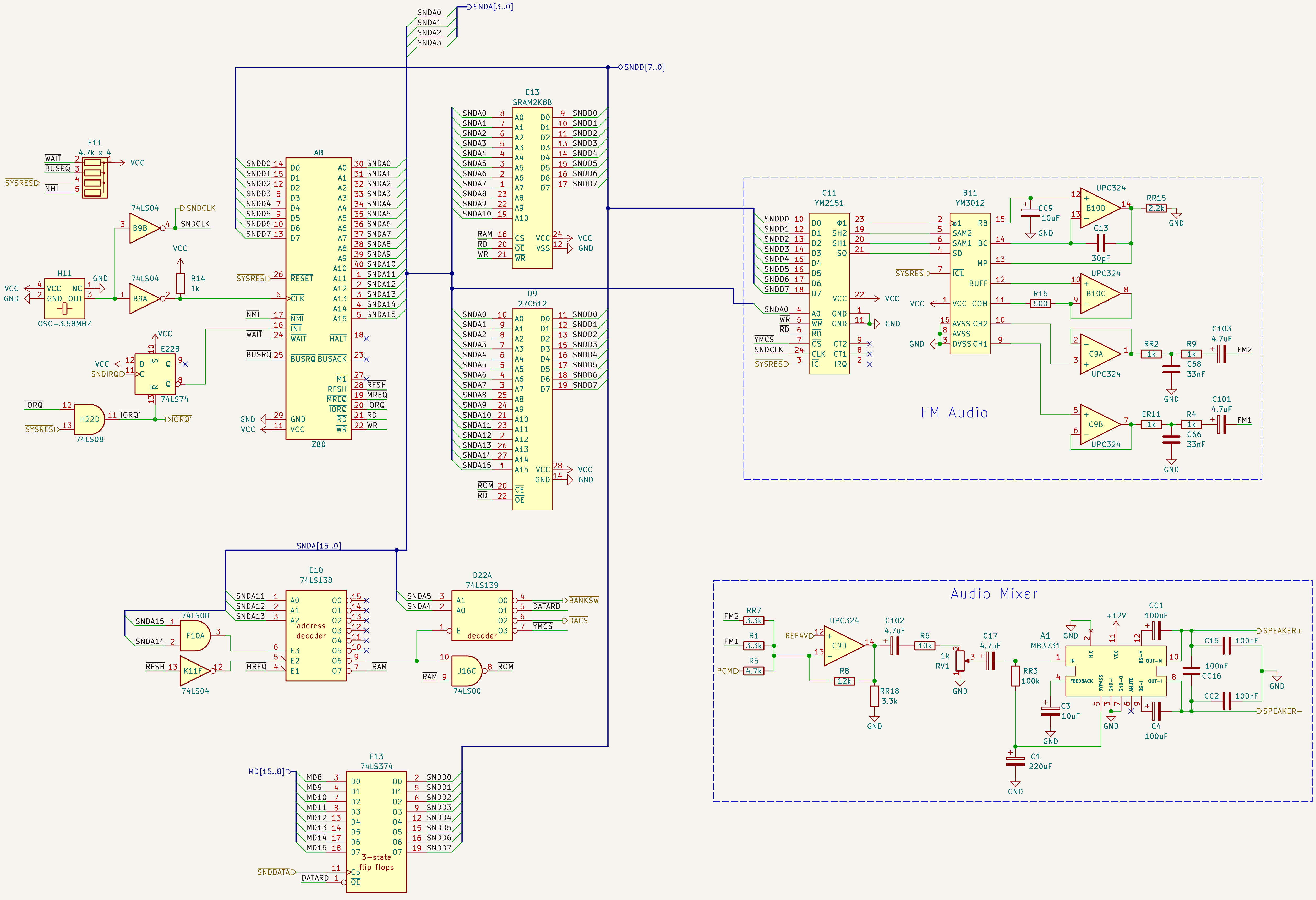
Sheet: /Color Mixer/

File: color_mixer.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	Rev:
KiCad E.D.A. 8.0.9		Id: 8/12



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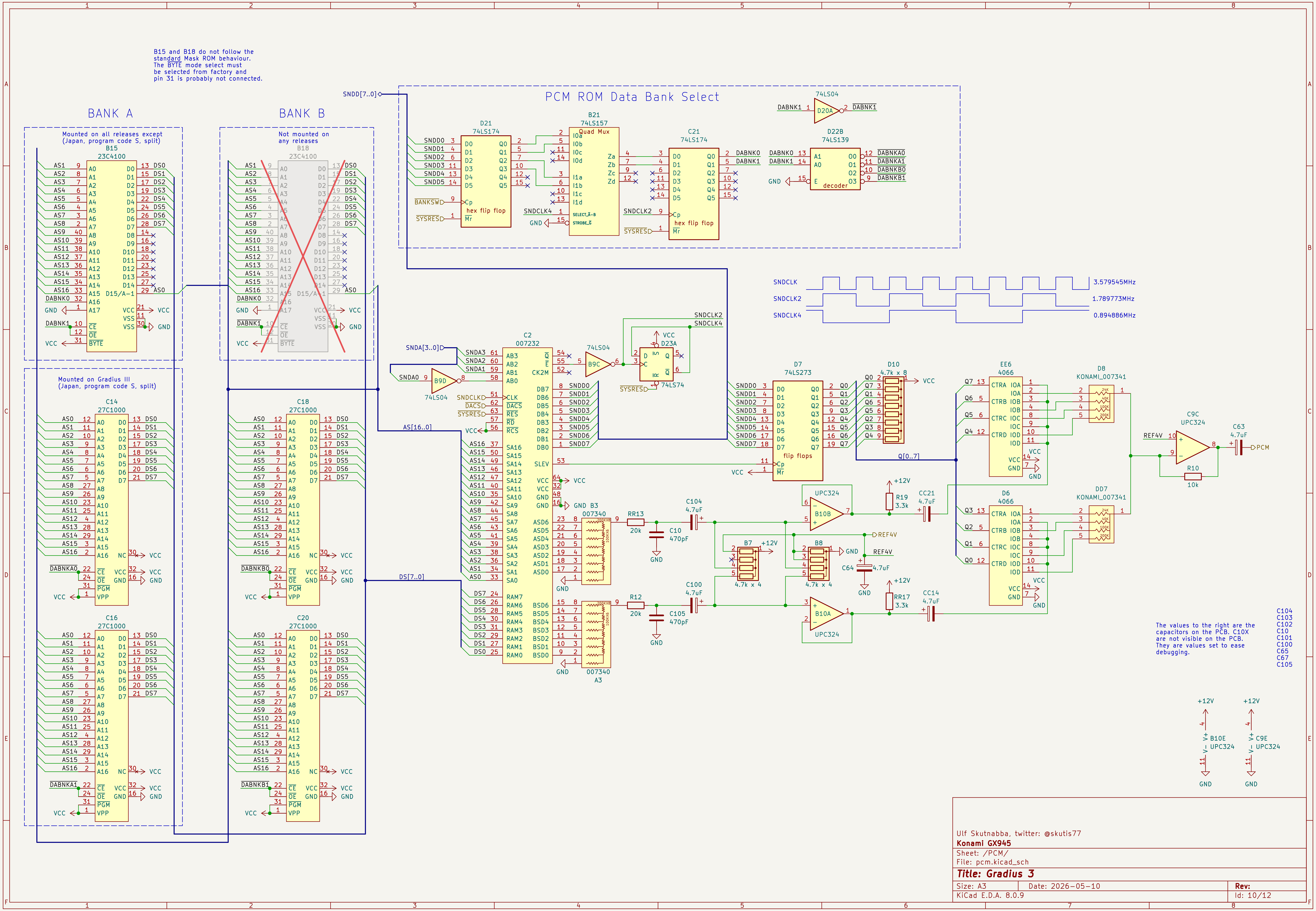
Sheet: /Sound/

File: sound.kicad_sch

Title: Gradius 3

Size: A3	Date: 2026-05-10	Rev:
KiCad E.D.A. 8.0.9		Id: 9/12


B15 and B18 do not follow the
standard Mask ROM behaviour.
The BYTE mode select must
be selected from factory and
pin 31 is probably not connected.
BANK A
Mounted on all releases except
(Japan, program code S, split)
B15
23C4100
BANK B
Not mounted on
any releases
B18
23C4100
Mounted on Gradius III
(Japan, program code S, split)
C14
27C1000
C16
27C1000
C18
27C1000
C20
27C1000
PCM ROM Data Bank Select
D21
74LS174
hex flip flop
B21
74LS157
Quad Mux
C21
74LS174
hex flip flop
74LS04
D20A
D22B
74LS139
decoder
SNDCLK
SNDCLK2
SNDCLK4
3.579545MHz
1.789773MHz
0.894886MHz
C2
007232
D23A
74LS74
D7
74LS273
flip flops
D10
4.7k x 8
EE6
4066
D8
KONAMI_007341
D6
4066
DD7
KONAMI_007341
C9C
UPC324
C63
4.7uF
PCM
REF4V
R10
10k
B3
007340
A3
007340
RR13
20k
R12
20k
C10
470pF
C105
470pF
C104
4.7uF
C100
4.7uF
B7
4.7k x 4
B8
4.7k x 4
C64
4.7uF
B10B
UPC324
B10A
UPC324
R19
3.3k
RR17
3.3k
CC21
4.7uF
CC14
4.7uF
The values to the right are the
capacitors on the PCB. C10X
are not visible on the PCB.
They are values set to ease
debugging.
C104
C103
C102
C10
C101
C100
C65
C67
C105
B10E
UPC324
C9E
UPC324
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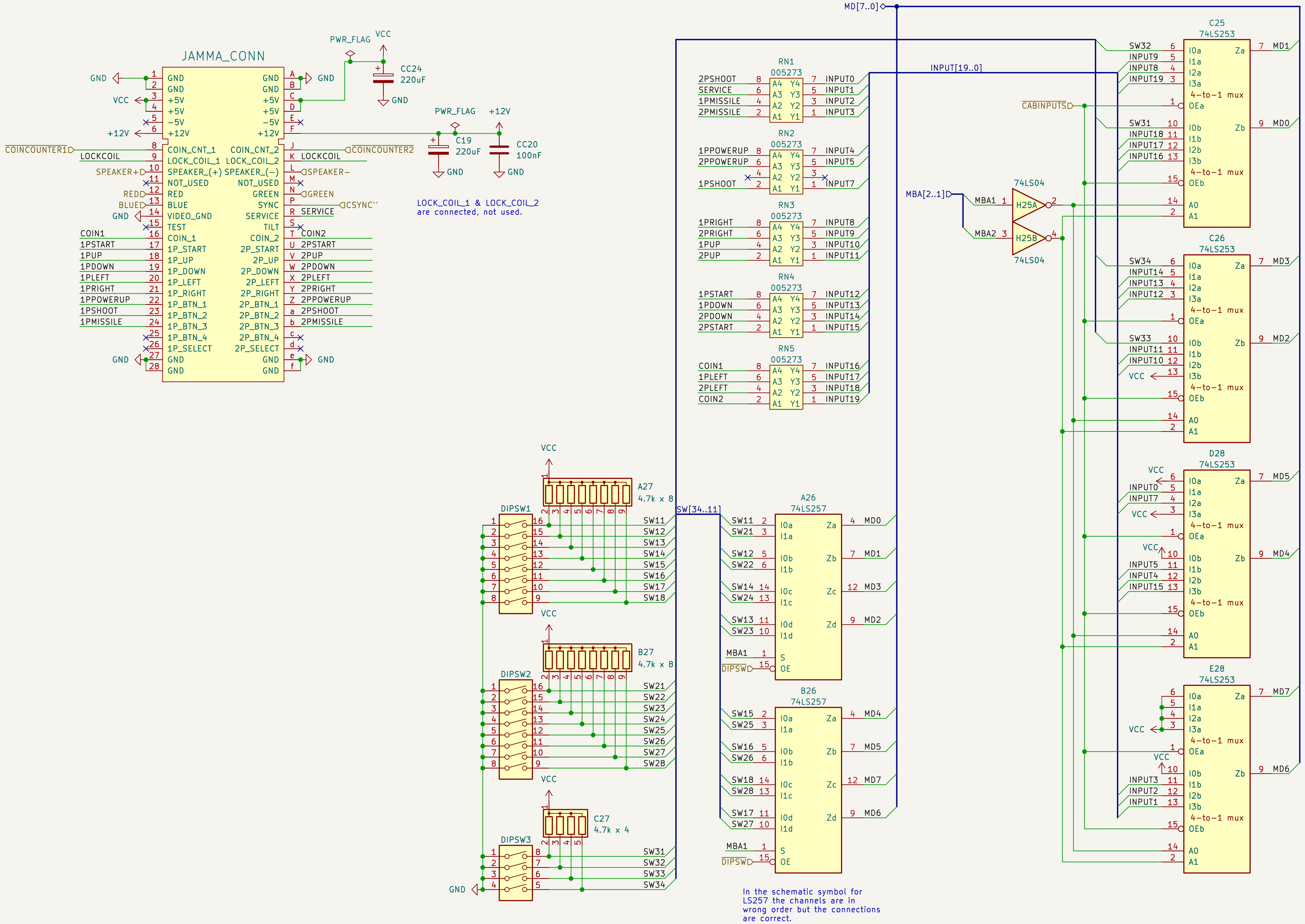
Konami GX945
Sheet: /PCM/
File: pcm.kicad_sch
Title: Gradius 3
Size: A3
Date: 2026-05-10
Rev:
KiCad E.D.A. 8.0.9
Id: 10/12



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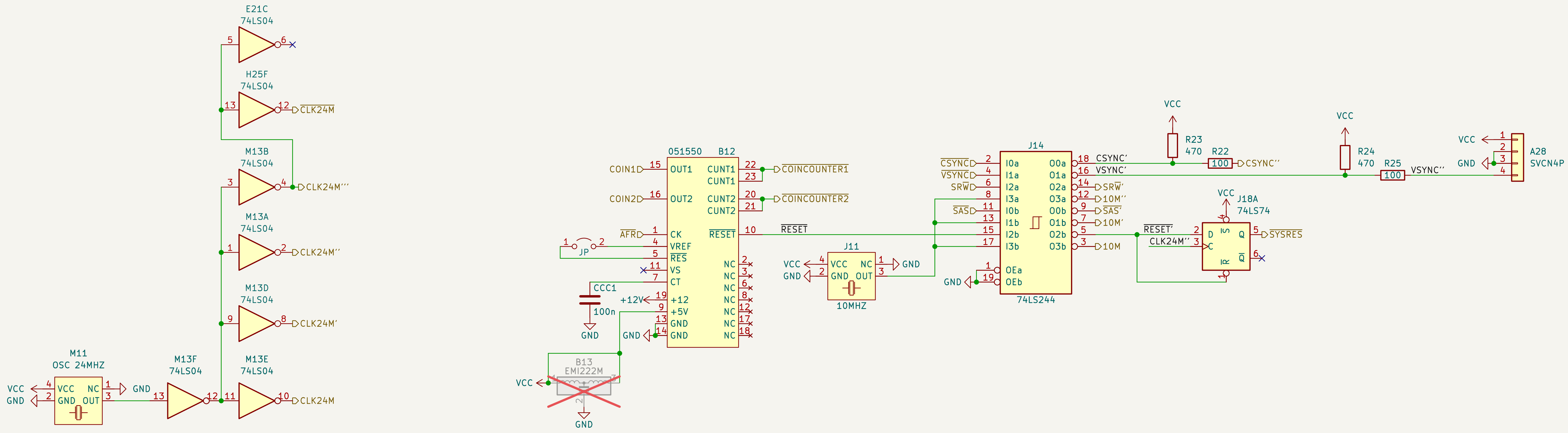
Sheet: /IO/

File: io.kicad_sch

Title: Gradius 3

Size: A3 | Date: 2026-05-10 | **Rev:**

KiCad E.D.A. 8.0.9 | Id: 11/12



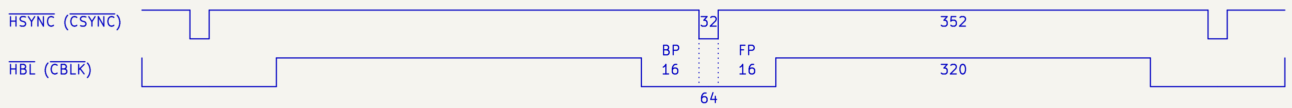
Horizontal and vertical synch timing diagrams

The pixel clock is derived from the 24MHz oscillator.
Pixel clock OVCK: f = 24MHz / 4 = 6MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of the pixel clock.

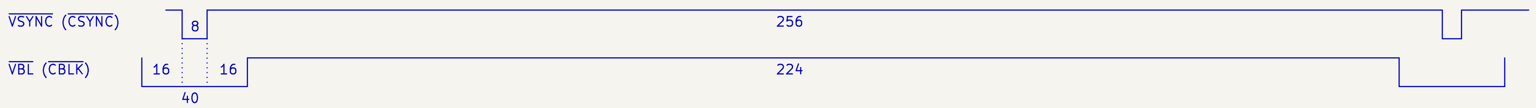
The signals have been measured at the output of the graphic chips.

If horizontal blanking is measured at the RGB DACs, the blanking is delayed 2 pixel clocks relative to composite sync. This gives BP = 14 and FP = 18.



HSYNC and HBL
Frequency f = 6MHz / 384 = 15.625kHz.
Period T = 1/f = 64us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.625kHz / 264 = 59.1856Hz
Period T = 1/f = 1 / 59.1856Hz = 16.896ms

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Konami GX945
Sheet: /Misc/
File: misc.kicad_sch
Title: Gradius 3
Size: A3 | Date: 2026-05-10 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 12/12